



Exercises session 4

P and NP

1 Definitions

Definition 1.1 (Verifier) A **verifier** for a language L is a deterministic Turing machine V such that

$$L = \{w \mid V \text{ accepts } \langle w, c \rangle \text{ for some string } c\}$$

A **polynomial time verifier** V runs in time that is polynomial in w . A language is **polynomially verifiable** if it has a polynomial time verifier.

Definition 1.2 (NTIME)

$$NTIME(t(n)) = \{L \mid L \text{ is a language decided by a } O(t(n)) \text{ time non-deterministic TM}\}$$

Definition 1.3 (Class NP) Let's see definitions of the class NP.

1. NP is the class of all languages that have polynomial time verifiers.
2. Defining NP via NTMs

$$NP = \bigcup_k NTIME(n^k)$$

Theorem 1 A language is in NP (has a poly-time verifier) if and only if it is decided by some nondeterministic Turing machine running in polynomial time.

Definition 1.4 (Kleene star operation) Let A be a language. The operation **star** $(*)$ is defined as follows:

$$A^* = \{x_1x_2 \dots x_k \mid k \geq 0 \text{ and each } x_i \in A \text{ where } 1 \leq i \leq k\}$$

2 Session exercises

1. Prove that the class **NP** is closed under union, intersection, kleene star and concatenation.
2. Let's G be an undirected graph. A path is considered simple if it doesn't contain repeated nodes.

- Prove that

$\text{SPATH} = \{\langle G, a, b, k \rangle : \text{there's a simple path from } a \text{ to } b \text{ of length at most } k\}$

is in **P**

- Is the following language also in **P**?

$\text{LPATH} = \{\langle G, a, b, k \rangle : \text{there's a simple path from } a \text{ to } b \text{ of length at least } k\}$

- Prove that **LPATH** is in **NP**

3. Let's introduce the SUBSET-SUM problem: given a set of integers $\mathcal{A} = \{a_1, \dots, a_n\}$ and another integer K represented in binary. Is there a subset of $\mathcal{A}$ that sums exactly K ?

- Is this language in **P**?
- Now let's think of the same problem, but the integers are represented in unary. How does our complexity analysis change?
- Prove that this problem is in **NP**

4. Given a graph $G = (V, E)$, an **independent set** of G is a set $X \subseteq V$ such that for all $(u, v) \in X \times X$, $(u, v) \notin E$.

- Give an example of a graph and two non-empty sets of vertices, such that one is an independent set and the other is not
- Prove that the following problem is in **NP**

$L = \{\langle G, k \rangle \mid G \text{ has an independent set of size } k\}$